Juanita Benjamin

Ph.D student— Computer Science

Nationality: US Citizen

juanitab@vt.edu linkedin.com juanita.com

Education

Ph.D., Computer Science **2022–Present**

Virginia Tech (VT) *Blacksburg, VA*

- Advisor: Ryan p. McMahan
- Continuing doctoral studies after transfer from University of Central Florida

Master of Science in Computer Science **2023–2025**

University of Central Florida (UCF) *Orlando, FL*

Bachelor of Science in Computer Science **2018–2022**

Texas Tech University (TTU) *Lubbock, TX*

Research Summary

XR/VR Engineer and HCI Researcher with 5+ years of experience designing and building immersive systems in Unity and C#. Proven track record of delivering interactive prototypes, running large-scale user studies, and analyzing multimodal datasets for insights. Published in top venues, including **IEEE VR**, **ISMAR**, and **IEEE TVCG**. Skilled in building end-to-end pipelines, experiment automation, and model evaluation in Python. Passionate about building immersive experiences and translating research into real-world applications.

Key Skills

- **Programming:** C#, Python, Unity, Git
- **XR/Software:** OpenXR, Meta SDK, XR Interraction Toolkit, VR prototyping
- **Data/ML:** Scikit-learn, pandas, NumPy, model training & evaluation
- **Research:** Experiment design, User studies, Peer-reviewed research, Collaboration
- **Domains:** XR/VR, AR, HCI, Data Analysis, Embodiment, Interactive Design

Publications

Journal

1. Tiffany. D.Do, **Juanita Benjamin**, Camille I. Protko and Ryan. P. McMahan, "Cultural Reflections in Virtual Reality: The Effects of User Ethnicity in Avatar Matching Experiences on Sense of Embodiment," *in IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 11, pp. 7408-7418, Nov. 2024, doi:10.1109/TVCG.2024.3456196. Acceptance rate: 16.9%

Conference

1. **Juanita Benjamin**, Austin Erickson, Matthew Gottsacker, Gerd Bruder and Greg F. Welch, "Evaluating Transitive Perceptual Effects Between Virtual Entities in Outdoor Augmented Reality," 2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR), Orlando, FL, USA, 2024, pp. 619-629, doi:10.1109/VR58804.2024.00082. Acceptance rate: 16.1%
2. **Juanita. Benjamin**, Gerd Bruder, Carsten Neumann, Dirk Reiners, Carolina Cruz-Neira, and Greg F. Welch, "Perception and Proxemics with Virtual Humans on Transparent Display Installations in Augmented Reality", *IEEE International Symposium on Mixed and Augmented Reality (ISMAR), Sydney, Australia, 2023*. pp. 386-395, doi:10.1109/ISMAR59233.2023.00053. Acceptance rate: 20.4%

3. Jake Gonzalez, Chau Pham, Afamefuna Umejiaku, **Juanita Benjamin**, and Tommy Dang. 2021. "SkeletonVR: Educating Human Anatomy Through An Interactive Puzzle Assembly". *In The 12th International Conference on Advances in Information Technology (IAIT2021). Association for Computing Machinery*, New York, NY, USA, Article 36, 1–5. DOI:doi.org/10.1145/3468784.3471605
4. **Juanita Benjamin**, Tommy Dang. 2021. "Virtual Simulation of 3D Modeling". *In the 18th International Conference on Modeling, Simulation and Visualization Methods (MSV2021).* Computer Science, computer Engineering & Applied Computing, Forthcoming.

Workshop

1. Priscilla Ramos; Meelad Doroodchi; Austin Erickson; Hiroshi Furuya; **Juanita Benjamin**; Gerd Bruder; Gregory Welch, "Effects of Optical See-Through Displays on Self-Avatar Appearance in Augmented Reality," *2022 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, Singapore, Singapore, 2022, pp. 352-356, doi:10.1109/ISMAR-Adjunct57072.2022.00077. Acceptance rate: 21%

Professional Experience

Graduate Research Assistant / Lab Manager **August 2025 - Present**
Virginia Tech *Blacksburg, VA*

- Designing and developing VR interaction systems in Unity, enabling controlled experimentation on embodiment, perception, and user behavior.
- Building a machine learning pipeline to predict user traits using multimodal datasets from VR interaction logs and movement patterns.
- Implementing feature extraction, normalization, model training (kNN, RF, BGM), and automated evaluation using StratifiedGroupKFold.
- Leading end-to-end research pipeline: prototype development, data collection, statistical analysis, and publication.
- Building and maintaining lab infrastructure, including hardware inventory systems, study automation tools, and scheduling workflows.

Graduate Research Assistant **August 2022 - 2025**
University of Central Florida *Orlando, FL*

- Engineered multiple VR and AR prototypes to investigate human-avatar interaction, integrating Unity, C#, and spatial computing.
- Designed moderate to large-scale human subject studies, implemented logging/telemetry systems, and analyzed behavioral datasets to extract actionable findings.
- Worked with faculty and collaborators to refine prototypes into publishable research artifacts presented at IEEE VR, ISMAR, and TVCG.

Graduate Research Assistant **June 2024 - August 2024**
DARPA Perceptually-enabled Task Guidance (PTG) Program *Orlando, FL*

- Collaborated with interdisciplinary teams to support prototype deployment and evaluation.
- Optimized visual and interaction cues in Unity, improving training efficiency and reducing cognitive load.

Undergraduate Research Scholar **August 2019 - May 2022**
Texas Tech University *Lubbock, TX*

- Built 3D modeling simulations and visualization tools for aircraft and engine systems.
- Contributed to publishing research and lab demonstrations for military partners.

Junior XR Developer Intern **April 2021 - June 2021**
Unity College *Remote*

- Tested a VR curriculum application for marine biology coursework.
- Collaborated using Plastic SCM to streamline workflow.

Other Experience

Graduate Teaching Assistant **2024-2025**
University of Central Florida *Orlando, FL*

Awards

Dean's Assistantship Award **2025–2027**
Virginia Tech

McKnight Doctoral Fellowship **2022–2025**
University of Central Florida

Graduate Presentation Fellowship **2023**
University of Central Florida

Christopher Rodriguez Research Presentation Award **2021**
Texas Tech University

Benjamin A. Gilman Scholarship **2021**
Gilman Scholarship Program

Teaching Experience

- Foundations of HCI, Grader, UCF, 2024
- COP 2500 Concepts in Computer Science, UCF, 2025

Professional Service & Outreach

Conference Peer Review

- External Reviewer, *International Conference of Extended Reality, (ICXR)*, 2025
- Reviewer, *ACM Symposium on Virtual Reality Software and Technology (VRST)*, 2025
- Reviewer, *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2025, 2024
- Reviewer, *IEEE Conference on Virtual Reality and 3D User Interfaces, (VR)*, 2024
- External Reviewer, *Eurographics*, 2025

Outreach

- Membership Chair, *Girls Who Code*, 2023
- Vice-President, *Girls Who Code*, 2024

Professional Membership

- Student Member, *Institute of Electrical and Electronics Engineers (IEEE)*, 2022-2025